University of Maryland at College Park
Dept. of Aerospace Engineering

ENAE 641: Linear System Dynamics

Problem Set #2

Issued: 3 Oct. 2026
Due: 13 Oct. 2026

Question 1:

For a time-invariant linear system with:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 0 & -13 \\ 1 & 0 & -17 \\ 0 & 1 & -5 \end{bmatrix} \qquad \boldsymbol{B} = \begin{bmatrix} 13 & 1 \\ 4 & 1 \\ 1 & 0 \end{bmatrix} \qquad \boldsymbol{C} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

and $\boldsymbol{D} = \mathbf{0}$:

a.) Determine the representation of the state dynamics in the real, block-modal form, and identify the required coordinate transformation matrix $\boldsymbol{P}$.

b.) Give a physically meaningful (i.e. real valued) initial state $\mathbf{x}_0$ which will excite **only** the oscillatory mode of the homogeneous state dynamics. Repeat for the purely exponential mode.

c.) Explicitly compute the transfer function matrix $\boldsymbol{G}(s)$ using the transformed state-space model in a), and verify that the result is identical to that obtained using the original state-space model. Matlab may be used for this; the commands `ss` and `tf` may be helpful.

Question 2:

The homogeneous state equation for a linear system has matrix $\boldsymbol{A}(t)$ that can be partitioned as

$$\begin{bmatrix} \boldsymbol{A}_1(t) & \boldsymbol{A}_{1,2}(t) \\ \mathbf{0} & \boldsymbol{A}_2(t) \end{bmatrix}$$

$\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are square, but of possibly different sizes, while $\boldsymbol{A}_{1,2}$ is not necessarily square.

a.) Show that the state transition matrix has the compatible form:

$$\boldsymbol{\Phi}(t,\tau) = \begin{bmatrix} \boldsymbol{\Phi}_1(t,\tau) & \boldsymbol{\Phi}_{1,2}(t,\tau) \\ \mathbf{0} & \boldsymbol{\Phi}_2(t,\tau) \end{bmatrix}$$

where $\boldsymbol{\Phi}_1$ is the STM for $\boldsymbol{A}_1(t)$ and $\boldsymbol{\Phi}_2$ is the STM for $\boldsymbol{A}_2(t)$

b.) Give an expression for $\boldsymbol{\Phi}_{1,2}(t,\tau)$ in terms of $\boldsymbol{\Phi}_1$, $\boldsymbol{\Phi}_2$, and $\boldsymbol{A}_{1,2}$. Note: $\boldsymbol{\Phi}_{1,2}$ will not be a STM for $\boldsymbol{A}_{1,2}$, since that matrix is not necessarily square.

Question 3: Given the state-space model:

$$\begin{aligned} \dot{x}_1(t) &= t x_2(t) \\ \dot{x}_2(t) &= -x_2(t) + e^{-t}u(t) \\ y(t) &= t x_1(t) \end{aligned}$$

a.) Find the state transition matrix $\Phi(t, \tau)$.

b.) Find the impulse response $G(t, \tau)$.

b.) Find the response $y(t)$ for $t \geq 1$ when $\mathbf{x}(1)^T = [1 \quad -2]$ and $u(t) = 1$ for all $t \geq 1$.

Question 4:

The characteristic polynomial of the $\boldsymbol{A}$ matrix for an LTI dynamic system can be written as $r(\lambda) = (\lambda^2 + 2\lambda + 10)(\lambda + 3)^3$. You also know that the rank of $(\boldsymbol{A} + 3\boldsymbol{I})$ is 4.

Let $\boldsymbol{P}$ be the matrix which will transform $\boldsymbol{A}$ to an appropriate real block modal form. Assume the $\boldsymbol{P}$ matrix is such that the components of the modal form arising from the complex eigenvalues will be located in the upper left corner.

a.) Compute the matrix $\boldsymbol{P}^{-1} e^{\boldsymbol{A}t} \boldsymbol{P}$.

b.) Suppose that we identify the outputs of this system to be $\mathbf{y}(t) = \boldsymbol{P}^{-1}\mathbf{x}(t)$. If the initial condition is $\mathbf{x}_0 = \mathbf{p}_4$ where $\mathbf{p}_4$ is the fourth column of the $\boldsymbol{P}$ matrix, find $\mathbf{y}(t)$ explicitly as a function of time.

c.) Suppose in b) that the initial condition is instead $\mathbf{x}_0 = \mathbf{p}_2$. Find $\mathbf{y}(t)$.

Question 5:

Suppose the matrix $\boldsymbol{A}(t)$ for a linear system can be written as

$$\boldsymbol{A}(t) = \sum_{j=1}^{r} f_j(t) \boldsymbol{A}_j$$

where the $\boldsymbol{A}_j$ are constant, and additionally $\boldsymbol{A}_i \boldsymbol{A}_j = \boldsymbol{A}_j \boldsymbol{A}_i$ (commutative) for every i, j.

a.) Show that the state transition matrix for this system can be written as a matrix exponential function.

b.) Show that the matrix exponential form for the STM can in fact be factored as

$$\boldsymbol{\Phi}(t, \tau) = e^{\boldsymbol{A}_1 \int_\tau^t f_1(\sigma) d\sigma} e^{\boldsymbol{A}_2 \int_\tau^t f_2(\sigma) d\sigma} \cdots e^{\boldsymbol{A}_r \int_\tau^t f_r(\sigma) d\sigma}$$

c.) Use the observations above to compute the state transition matrix for

$$\boldsymbol{A}(t) = \begin{bmatrix} \cos 3t & \sin 3t \\ -\sin 3t & \cos 3t \end{bmatrix}$$

Question 6:

For the homogeneous linear system $\dot{\mathbf{x}}(t) = \boldsymbol{A}(t)\mathbf{x}(t)$, find the state transition matrix $\boldsymbol{\Phi}(t, \tau)$ when:

a.) $\boldsymbol{A}(t) = \begin{bmatrix} t & 0 \\ 0 & \cos(t) \end{bmatrix}$ b.) $\boldsymbol{A}(t) = \begin{bmatrix} \cos(2t) & \cos(2t) \\ 0 & 0 \end{bmatrix}$

c.) $\boldsymbol{A}(t) = \begin{bmatrix} -2 & \sin(t) \\ 0 & -1 \end{bmatrix}$ d.) $\boldsymbol{A}(t) = \begin{bmatrix} -3 & -1 \\ -1 & -3 \end{bmatrix}$

Show all work for your solutions. In d), use a dyadic expansion approach.

Question 7:

Consider a time-periodic system with

$$\boldsymbol{A}(t) = \begin{bmatrix} -1 + 1.5\cos^2(t) & 1 - 1.5\sin(t)\cos(t) \\ -1 - 1.5\sin(t)\cos(t) & -1 + 1.5\sin^2(t) \end{bmatrix}$$

a.) Find the eigenvalues of $\boldsymbol{A}(t)$, show that they do not change with time, and are located in the left-half of the complex plane.

b.) Verify that two fundamental solutions for this system are:

$$\boldsymbol{\psi}_1(t) = \begin{bmatrix} e^{-t}\sin(t) \\ e^{-t}\cos(t) \end{bmatrix} \qquad \boldsymbol{\psi}_2(t) = \begin{bmatrix} e^{t/2}\cos(t) \\ -e^{t/2}\sin(t) \end{bmatrix}$$

c.) Determine the state transition matrix for this system. Is this system stable; i.e. does $\mathbf{x}(t) \to \mathbf{0}$ from every initial condition $\mathbf{x}_0$?

d.) Comparing your results in a) and c), comment on the utility of the eigenvalues of a time-varying $\boldsymbol{A}(t)$ for analyzing stability.